Department of Mathematics, IIT Madras

MA 2102 - Differential Equations, ODD Semester, 2026-2027

Assignment-10 : Laplace Equations

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1. Write the Laplace equation $u_{xx} + u_{yy} = 0$ in polar coordinates.

2. Solve the following Laplace equations:

   (a) $\nabla^2 u = 0;\ 0 < x < 1,\ 0 < y < 1,$
   $u(x,0) = 0, u(x,1) = \sin(\pi x), 0 \le x \le 1,$
   $u(0,y) = 0, u(1,y) = 0, 0 \le y \le 1.$

   (b) $\nabla^2 u = 0;\ 0 < x < 1,\ 0 < y < 1,$
   $u(x,0) = x(1-x), u(x,1) = 0, 0 \le x \le 1,$
   $u(0,y) = y(1-y), u(1,y) = 0, 0 \le y \le 1.$

   (c) $\nabla^2 u = 0;\ 0 \le r \le a,\ 0 \le \theta \le 2\pi,$
   $u(a,\theta) = f(\theta).$

   (d) $\nabla^2 u = 0;\ 1 < r < 2,\ 0 < \theta < \pi,$
   $u(1,\theta) = \sin\theta, u(2,\theta) = 0, 0 \le \theta \le \pi,$
   $u(r,0) = 0, u(r,\pi) = 0, 1 \le r \le 2.$

3. A square plate of side π units occupying the region $0 \le x \le \pi$, $0 \le y \le \pi$ has its edges $x = 0$, $x = \pi$, $y = \pi$ insulated. The side $y = 0$ is kept at temperature $u(x,0) = x$, $0 < x < \pi$. Find the steady state temperature distribution in the plate.

   Ans:
   $$u(x,y) = \frac{c_0}{2} + \sum_{n=1}^{\infty} c_n \cos nx \cosh n(y-\pi)$$
   where
   $$c_0 = \pi, \qquad c_n = \frac{2}{n^2\pi \cosh n\pi}\big[\cos n\pi - 1\big], \qquad n = 1,2,3,\ldots$$

4. A thin plate occupying the region $0 \le x \le 1$, $0 \le y \le 1$ has the edges $y = 0$ and $y = 1$ maintained at 0^oc. The edge $x = 0$ is maintained at the temperature $y(1-y)$, $0 < y < 1$. The edge $x = 1$ is insulated. Find the steady state temperature distribution in the plate.

   Ans:
   $$u(x,y) = \sum_{n=1}^{\infty} c_n \cosh n\pi(x-1) \sin(n\pi y)$$
   where
   $$c_n = \frac{4}{n^3\pi^3 \cosh n\pi}\big[1 - (-1)^n\big], \qquad n = 1,2,3,\ldots$$

5. Solve

$$u_{xx} + u_{yy} = 0, \quad 0 < x < \pi,\ 0 < y < \pi,$$

$$u(0, y) = u(\pi, y) = 0, \quad 0 < y < \pi,$$

$$u(x, \pi) = 0, \quad 0 < x < \pi,$$

$$u(x, 0) = \sin^2(x), \quad 0 < x < \pi.$$

Ans:

$$u(x, y) = \sum_{n=1}^{\infty} \frac{4\big[(-1)^n - 1\big]}{n\pi(n^2 - 4)\sinh n\pi} \sinh\big(n(\pi - y)\big) \sin nx$$

6. A rectangular plate with insulated surface is 10 cm wide and so long compared to its width that it may be considered infinite in length without introducing appreciable error. If the temperature of the short edge $y = 0$ is given by

$$u(x, 0) = \begin{cases} 20x, & 0 \le x \le 5 \\ 20(10 - x), & 5 \le x \le 10 \end{cases}$$

and the two long edges $x = 0$, $x = 10$ as well as the other short edge are kept at 0^oc, find the steady state temperature distribution in the plate.

Ans:

$$u(x, y) = \sum_{n=1}^{\infty} \frac{800}{n^2\pi^2} \sin\Big(\frac{n\pi}{2}\Big) \sin\Big(\frac{n\pi x}{10}\Big) e^{-n\pi y/10}.$$

7. A plate is in the form of a semi-infinite strip $0 \le x \le L$, $0 \le y < \infty$. The edges $x = 0$ and $x = L$ are insulated and the edge $y = 0$ is kept at the temperature $2\cos\left(\dfrac{3\pi x}{L}\right) + 3\cos\left(\dfrac{4\pi x}{L}\right)$, $0 \le x \le L$. Find the steady state temperature $u(x, y)$ in the plate assuming that $\lim\limits_{y\to\infty} u(x, y) = 0$, $0 \le x \le L$.

Ans:

$$u(x, y) = 2\cos\left(\frac{3\pi x}{L}\right) e^{-3\pi y/L} + 3\cos\left(\frac{4\pi x}{L}\right) e^{-4\pi y/L}$$

8. Find u satisfying

$$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0, \quad 0 \le \theta \le 2\pi,\ r < 1,$$

subjected to the boundary condition

(a) $u(1, \theta) = 10\cos 2\theta,\ 0 \le \theta \le 2\pi.$

(b) $u(1, \theta) = \begin{cases} \theta, & -\dfrac{\pi}{2} \le \theta < \dfrac{\pi}{2} \\ 0, & \dfrac{\pi}{2} \le \theta < \dfrac{3\pi}{2}. \end{cases}$

Ans:

(a) $u(r, \theta) = 5(1 + r^2\cos 2\theta)$

(b) $u(r, \theta) = \sum\limits_{n=1}^{\infty} \dfrac{r^n}{2\pi n^2} \sin\Big(\dfrac{n\pi}{2}\Big) \sin n\theta$

9. Find u satisfying

$$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0, \quad 0 \le \theta \le \pi,\ r < 1,$$

subjected to the boundary conditions

$$u(r,0) = 0 = u(r,\pi), \qquad u(1,\theta) = 10\theta(\pi - \theta), \quad 0 \le \theta \le \pi.$$

Ans:

$$u(r,\theta) = \sum_{n=1}^{\infty} \frac{40}{\pi n^3}\big(1 - \cos n\pi\big) r^n \sin n\theta$$

10. $u(r,\theta)$ is a function satisfying

$$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0, \quad 0 \le \theta \le \frac{\pi}{2},\ a < r < b.$$

If its value along the boundary $r = a$ is $\theta\Big(\frac{\pi}{2} - \theta\Big)$ and is zero on the remaining part of the boundary, then prove that

$$u(r,\theta) = \frac{2}{\pi}\sum_{n=1}^{\infty} \frac{\left(\frac{r}{b}\right)^{4n-2} - \left(\frac{b}{r}\right)^{4n-2}}{\left(\frac{a}{b}\right)^{4n-2} - \left(\frac{b}{a}\right)^{4n-2}} \cdot \frac{\sin(4n-2)\theta}{(2n-1)^3}$$